Customer Buying Behaviour Analysis

1. Project Overview

This Project analyses customer Behaviour using Transactional data across various product categories.

2. Dataset Summary

- Rows:3900
- Columns:18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
  - Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python-Pandas

We started with data preparation and cleaning using Pandas:

- **Data Loading:** Imported the dataset using pandas.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping

● **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.

```
]: #Overviewing Data structure: How many colums, rows, DataTypes
   df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
df.describe()
```

]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

```
#All
df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discour Applie
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	390
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	N
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	222
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	Nal
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	Nal
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	Nal
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	Nal
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	Nal
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	Nal
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	Nal

- **Missing Data Handling:** Checked for null values and imputed missing values in the “Review Rating” column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
    - Created **age_group** column by binning customer ages.
    - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Gender**: Compared total revenue generated by male vs. female customers.

   #Q 01. What is the total revenue generated by male vs. female customers?

```
+--------+---------------+
| Gender | Total_Revenue |
+--------+---------------+
| Female |         75191 |
| Male   |        157890 |
+--------+---------------+
2 rows in set (0.03 sec)
```

2. **High-Spending Discount Users:** Identified customers who used discounts but still spent above the average purchase amount.

   #Q 02. Which customers used a discount but still spent more than the average purchase amount?

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
1654	93
1656	81
1659	66
1662	86
1667	64
1671	73
1673	73
1674	62
1676	90

839 rows in set (0.01 sec)

3. **Top 5 Products by Rating:** Found products with the highest average review ratings.
   #Q3. Which are the top 5 products with the highest average review rating?

item_purchased	avg_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

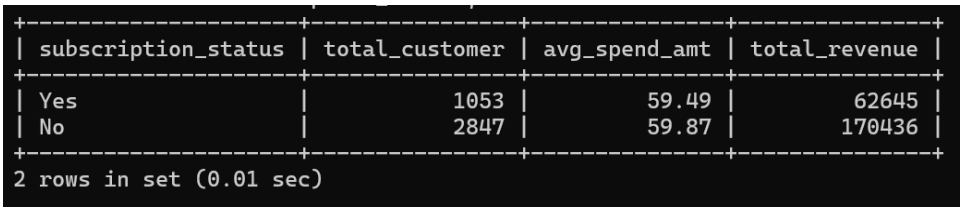
5 rows in set (0.02 sec)

4. **Shipping Type Comparison:** Compared average purchase amounts between Standard and Express shipping.
   #Q4. Compare the average Purchase Amounts between Standard and Express Shipping.

shipping_type	round(AVG(purchase_amount),2)
Express	60.48
Standard	58.46

2 rows in set (0.01 sec)

5. **Subscribers vs. Non-Subscribers:** Compared average spend and total revenue across subscription status.
   #Q5. Do subscribed customers spend more? Compare average spend and total revenue

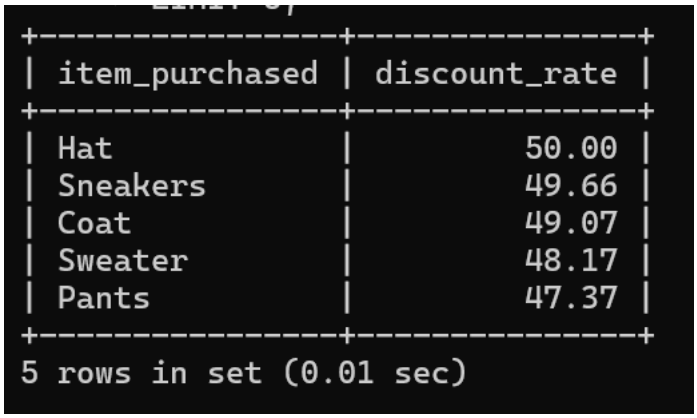


subscription_status	total_customer	avg_spend_amt	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

2 rows in set (0.01 sec)

6. **Discount-Dependent Products:** Identified 5 products with the highest percentage of discounted purchases.

   #Q6. Which 5 products have the highest percentage of purchases with discounts applied?



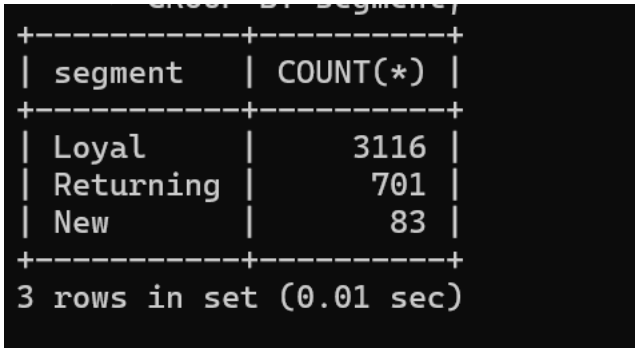
item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

5 rows in set (0.01 sec)

7. **Customer Segmentation:** Classified customers into New, Returning, and Loyal segments based on purchase history

   #Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

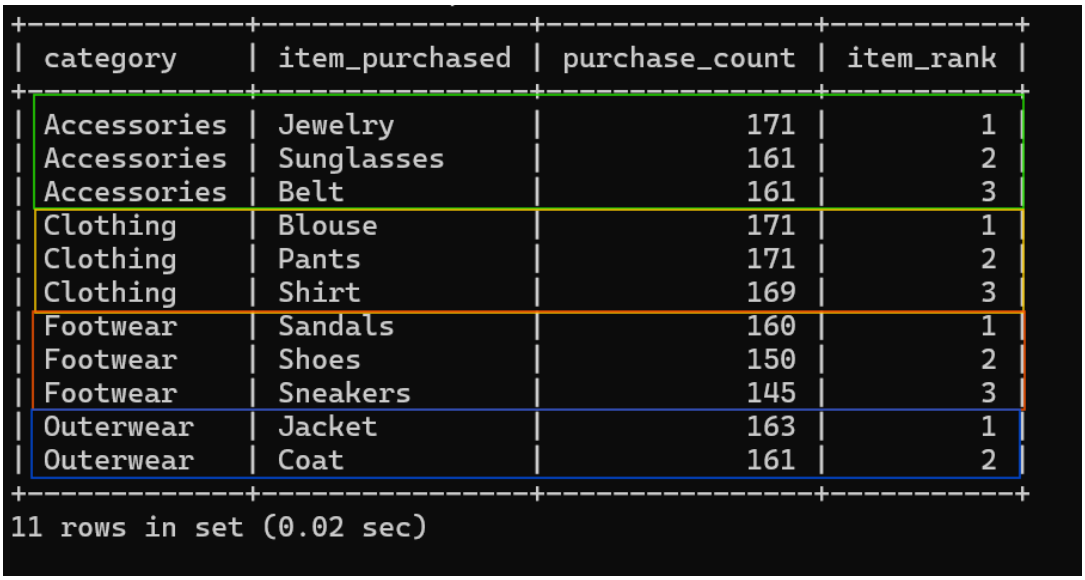
   # 1time = New; 2times - 10times = Returning; <10times = Loyal



segment	COUNT(*)
Loyal	3116
Returning	701
New	83

3 rows in set (0.01 sec)

8. **Top 3 Products per Category:** Listed the most purchased products within each category.
   #Q8. What are the top 3 most purchased products within each category?

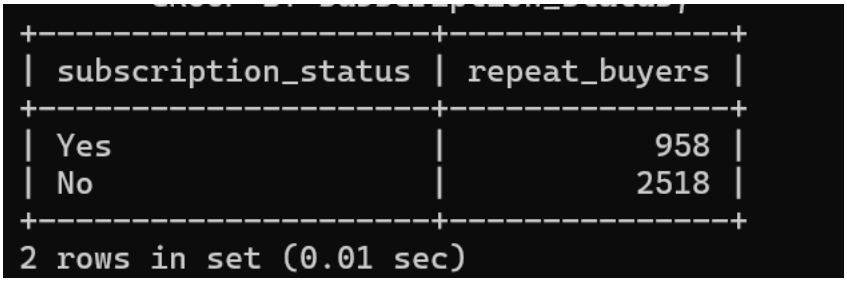


category	item_purchased	purchase_count	item_rank
Accessories	Jewelry	171	1
Accessories	Sunglasses	161	2
Accessories	Belt	161	3
Clothing	Blouse	171	1
Clothing	Pants	171	2
Clothing	Shirt	169	3
Footwear	Sandals	160	1
Footwear	Shoes	150	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

11 rows in set (0.02 sec)

9. **Repeat Buyers & Subscriptions:** Checked whether customers with >5 purchases are more likely to subscribe.

   #Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

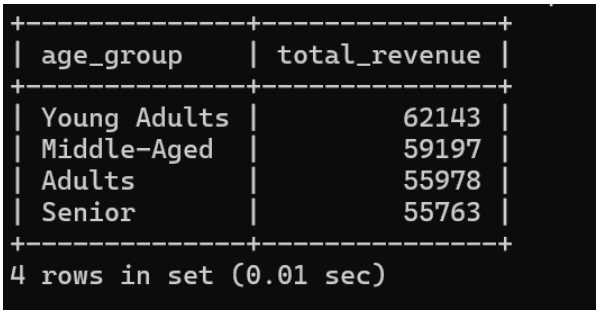


subscription_status	repeat_buyers
Yes	958
No	2518

2 rows in set (0.01 sec)

10. **Revenue by Age Group:** Calculated total revenue contribution of each age group.

    #Q10. What is the revenue contribution of each age group?



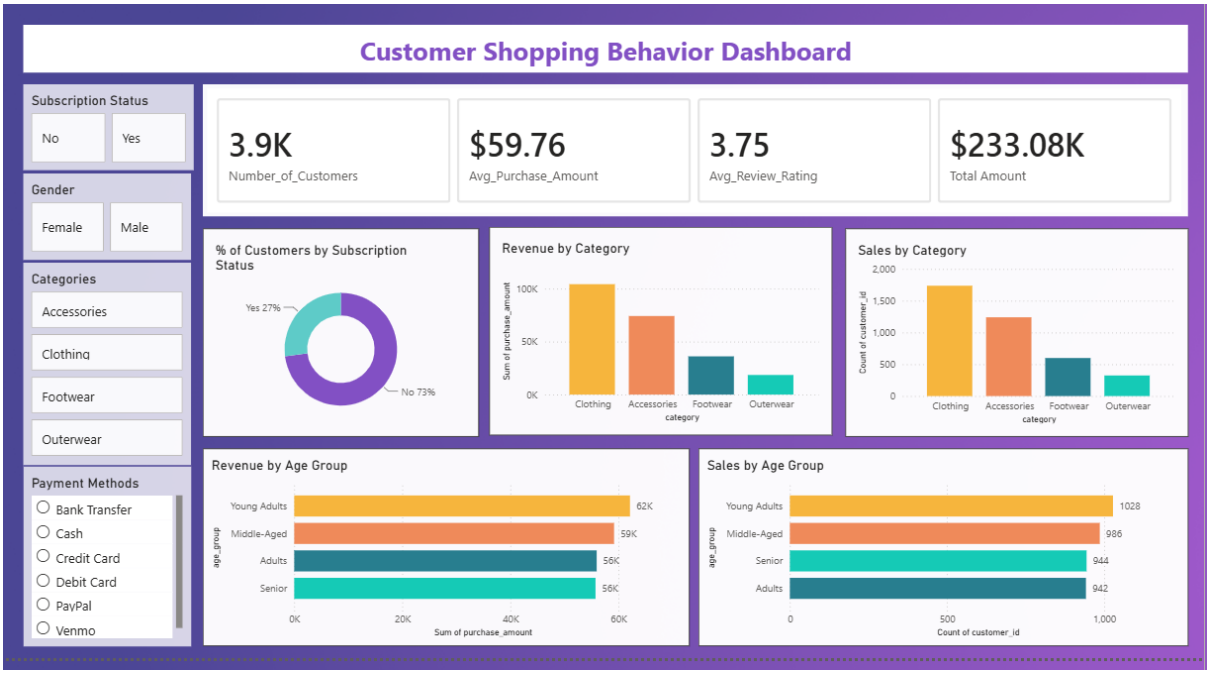
age_group	total_revenue
Young Adults	62143
Middle-Aged	59197
Adults	55978
Senior	55763

4 rows in set (0.01 sec)

5. Data Analysis using SQL (Business Transactions)

Finally, I built an interactive Power BI dashboard to present insights visually.

(As I come from the Fintech domain, I have added Payment Methods as a slicer to check the behaviour of customers' payment choices)



6. Business Recommendations

1. **Boost Subscriptions:** Promote exclusive benefits for subscribers so that customers can subscribe;
2. **Customer Loyalty Programs:** Reward repeat buyers to move them into the “Loyal” segment so that customers will buy more.
3. **Product Promotions:** Promote top-rated and best-selling products so that other customers buy those products too.
4. **Targeted Marketing:** Focus efforts on high-revenue age groups and express-shipping users.